

# Laravel To-Do List Application

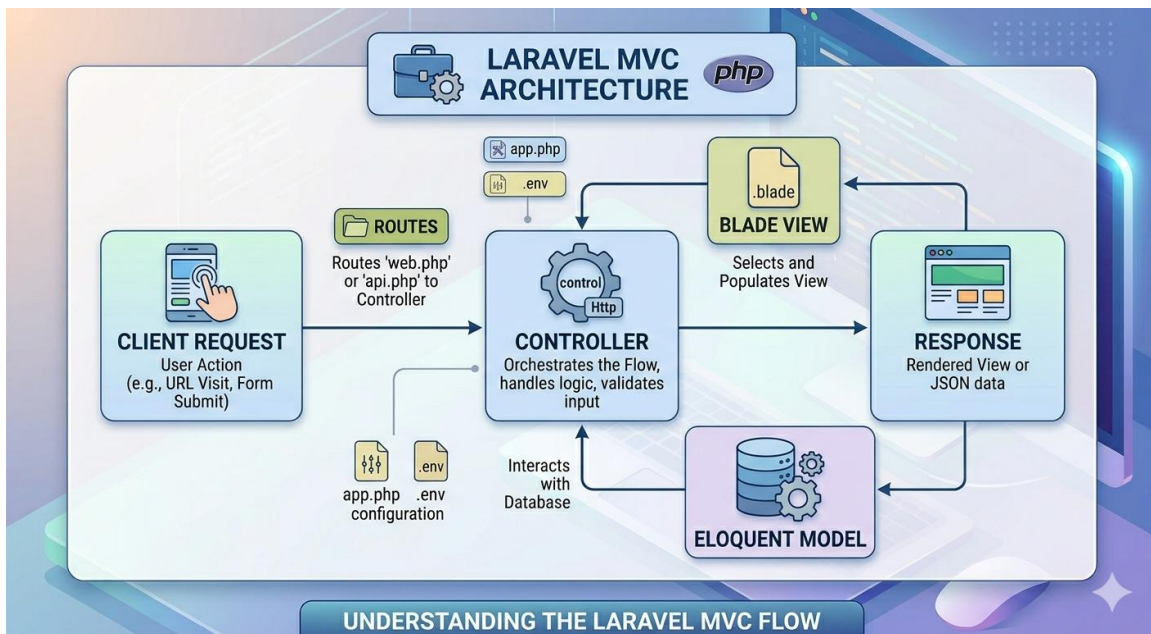
Laravel is a PHP framework used to build modern web applications easily and quickly.

It follows the MVC Architecture:

Model → Handles database (data)

View → What user sees (UI)

Controller → Logic (connects model + view)



Flow = User → Route → Controller → Model → Database → View → User

## **WHY THIS MATTERS?**

1. Keeps code organized
2. Easy to debug
3. Easy to scale

## **PROJECT SETUP**

Step 1: Install Laravel

```
composer create-project laravel/laravel todo-app
```

```
cd todo-app
```

```
php artisan serve
```

*composer create-project → downloads Laravel*

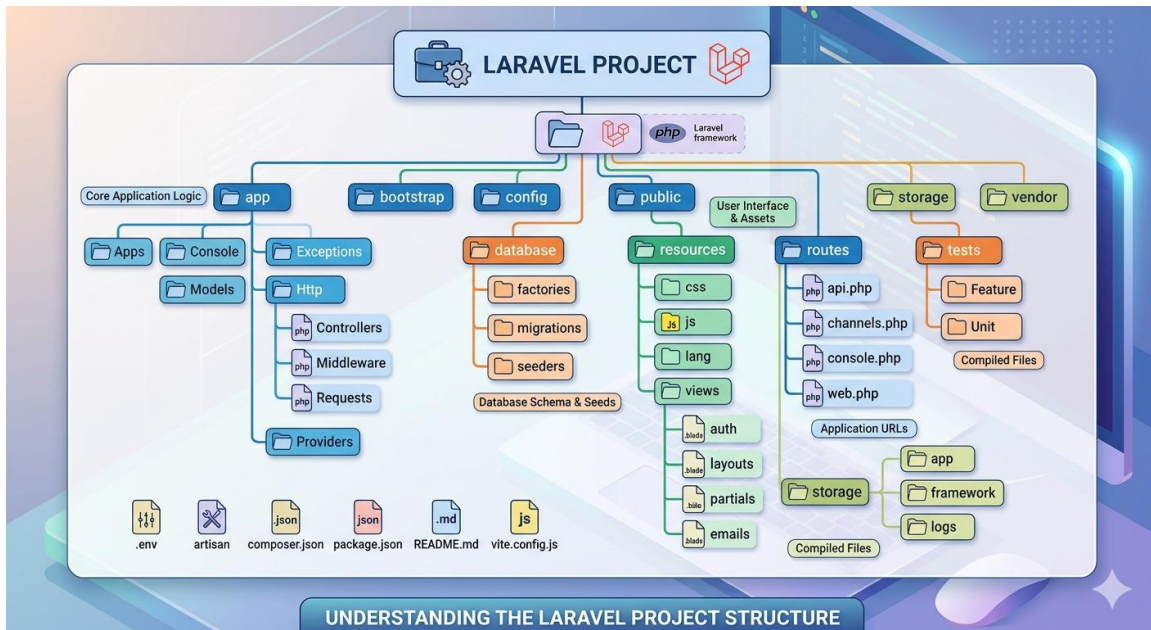
*cd → enter project*

*php artisan serve → runs server*

Open the browser and go to

*http://127.0.0.1:8000*

## Important Folder Structure



## DATABASE CONFIGURATION

### Step 1: Edit .env

`DB_DATABASE=todo_db`

`DB_USERNAME=root`

`DB_PASSWORD=`

### Create Database

#### In phpMyAdmin:

*Create database todo\_db;*

Why?

Laravel needs a database to **store tasks permanently**

## **MODEL & MIGRATION**

### **Run**

*Create Model + Migration*

*php artisan make:model Task -m*

### **MIGRATION CODE**

Should be inside directory : database/migrations/xxxx\_create\_tasks\_table.php

```
public function up()
{
    Schema::create('tasks', function (Blueprint $table) {
        $table->id(); // Primary key
        $table->string('title'); // Task name
        $table->boolean('completed')->default(false); // Status
        $table->timestamps(); // created_at & updated_at
    });
}
```

1.     \$table->id() → unique ID
2.     \$table->string('title') → task text
3.     \$table->boolean('completed') → true/false
4.     \$table->timestamps() → auto time

## RUN MIGRATION

*php artisan migrate*

This creates the tasks table

## MODEL EXPLANATION

Inside dir : app/Models/Task.php

```
protected $fillable = ['title', 'completed'];
```

## WHY?

Laravel protects database from unsafe data.

**fillable allows:**

1. safe insertion of data
2. prevents hacking (mass assignment issue)

## CONTROLLER (CORE LOGIC)

### Create Controller

Run on terminal :

```
php artisan make:controller TaskController
```

Go to dir : app/Http/Controllers/TaskController.php

**Change code to :**

```
use App\Models\Task;
```

```
use Illuminate\Http\Request;
```

```
class TaskController extends Controller
{
    // Show all tasks

    public function index()
    {
        $tasks = Task::all(); // get all tasks

        return view('tasks.index', compact('tasks'));
    }

    // Add task

    public function store(Request $request)
    {
        $request->validate([

            'title' => 'required'

        ]);

        Task::create([

            'title' => $request->title

        ]);

        return redirect('/');
    }
}
```

```
// Delete task

public function destroy($id)

{

    Task::findOrFail($id)->delete();

    return redirect('/');

}


// Mark complete

public function update($id)

{

    $task = Task::findOrFail($id);

    $task->completed = !$task->completed;

    $task->save();

    return redirect('/');

}

}
```

## ROUTING

File: routes/web.php

Code inside file :

```
use App\Http\Controllers\TaskController;

Route::get('/', [TaskController::class, 'index']);

Route::post('/add', [TaskController::class, 'store']);

Route::delete('/delete/{id}', [TaskController::class, 'destroy']);

Route::put('/update/{id}', [TaskController::class, 'update']);
```

## WHY?

Routes connect:

URL → Controller function

## VIEW (BLADE TEMPLATE)

Inside file you can use the code provided below but also you can create your own. This is just a reference. Go to :

**resources/views/tasks/index.blade.php**

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>To-Do App</title>
```



```
<style>

  body {

    font-family: Arial;

    background: #f4f4f4;

    display: flex;

    justify-content: center;

    margin-top: 50px;

  }

  .container {

    background: white;

    padding: 20px;

    width: 400px;

    border-radius: 10px;

    box-shadow: 0 0 10px gray;

  }

  .completed {

    text-decoration: line-through;

    color: gray;

  }

</style>

</head>
```

```
<body>
```

```

<div class="container">

    <h2>To-Do List</h2>

    <!-- Add Task -->

    <form action="/add" method="POST">

        @csrf

        <input type="text" name="title" placeholder="Enter task">

        <button>Add</button>

    </form>

    <!-- Task List -->

    @foreach($tasks as $task)

        <div>

            <span class="{{ $task->completed ? 'completed' : '' }}">

                {{ $task->title }}

            </span>

            <!-- Complete -->

            <form action="/update/{{ $task->id }}" method="POST">

                @csrf

                @method('PUT')

                <button>✓</button>

            </form>

            <!-- Delete -->

            <form action="/delete/{{ $task->id }}" method="POST">

                @csrf

```

```
        @method('DELETE')

        <button>✕</button>

    </form>

</div>

@endforeach

</div>

</body>

</html>
```

@csrf → security protection

@foreach → loop through tasks

{{ }} → display data

## APPLICATION FLOW

1. User enters task
2. Form sends data → Route
3. Route → Controller
4. Controller → Model
5. Model → Database
6. Data returns → View
7. User sees updated list

## TESTING

Run the app using

```
php artisan serve
```